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Chapter 1
Message Passing Schemes

Abstract Message Passing schemes are effective ways to evaluate contractions. We first introduce based on semirings a more generic definitions of tensor network contraction. We then show that any such contraction can be splitted based on hypergraph cuts, which enables message-passing schemes to efficiently perform contractions. In the special cases of Boolean and basis encoding tensor networks, we further show even more efficient algorithms.

1.1 Semiring contraction

We here introduce with commutative semirings the algebraic structure required for the main separability property of contractions, and therefore for message-passing algorithms.

See [6, 7] for applications and [1] for algebraic and graph-theoretic foundations.

In an even broader perspective, contractions can be defined based on sequential compositions of categories [**?**].

1.1.1 Foundations

Definition 1.1 A commutative semiring is a tuple $\mathbf{S} = (S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$, where S is a set, $\mathbf{0}, \mathbf{1} \in S$, and

$$\oplus : S \times S \to S \quad , \quad \otimes : S \times S \to S ,$$

such that the following properties hold:

(i) $(S, \oplus, \mathbf{0})$ is a commutative monoid[1].

(ii) $(S, \otimes, \mathbf{1})$ is a commutative monoid.

(iii) $\otimes$ distributes over $\oplus$, that is for any $s_0, s_1, s_2 \in S$

$$s_0 \otimes (s_1 \oplus s_2) = (s_0 \otimes s_1) \oplus (s_0 \otimes s_2) .$$

(iv) $\mathbf{0}$ annihilates $\otimes$, that is for any $s_0 \in S$

$$\mathbf{0} \otimes s_0 = \mathbf{0} .$$

Notice, that by these properties, the group operations $\oplus$ and $\otimes$ are specified on elements $\{0, 1\}$, except for $1 \oplus 1$, that is the following holds for any semiring:

$\oplus$	**0**	**1**
0	**0**	**1**
1	**1**	*

$\otimes$	**0**	**1**
0	**0**	**0**
1	**0**	**1**

[1] That is associativity, commutativity of $\oplus$ and identity of $s \oplus \mathbf{0} = s$ for all $s \in S$.

The Galois semiring and the Boolean semiring are the smallest semirings containing only the elements $\mathbf{0}, \mathbf{1}$ and differing only in $\mathbf{1} \oplus \mathbf{1}$.

We understand S as the set of possible coordinates of a tensor.

Definition 1.2 A tensor with coordinates in S of order $d \in \mathbb{N}$ and leg dimensions $m_{[d]} = (m_0, \dots, m_{d-1})$, where $m_k \in \mathbb{N}$ for $k \in [d]$, is a map

$$\tau : \bigtimes_{k \in [d]} [m_k] \to S .$$

A tensor network with coordinates in S is a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node is decorated by a dimension m_v and each edge by a tensor of order $|e|$ and leg dimensions m_e.

Definition 1.3 Given a semiring $\mathbf{S}$ and a tensor network $\{\tau^e [X_e] : e \in \mathcal{E}\}$ with coordinates in S, the contraction of the tensor network keeping the variables to a subset $\mathcal{U} \subset \mathcal{V}$ open is the tensor

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle^{\mathbf{S}}_{[X_{\mathcal{U}}]}$$

with coordinates at indices $x_{\mathcal{U}}$ by

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle^{\mathbf{S}}_{[X_{\mathcal{U}}]} = \bigoplus_{x_{\mathcal{V}\setminus\mathcal{U}}} \bigotimes_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

The distributive property (iii) of the semiring ensures, that we can compute contractions by sub-contractions.

Theorem 1.1 *Given a disjoint partition of $\mathcal{E}$ into $\mathcal{E}^0, \mathcal{E}^1$, and a subset $\mathcal{U} \subset \mathcal{V}$, we define the sets $\mathcal{V}^0$ and $\mathcal{V}^1$ as the nodes appearing in $\mathcal{E}^0$, respectively $\mathcal{E}^1$. We then have*

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle^{\mathbf{S}}_{[X_{\mathcal{U}}]} = \langle \langle \{\tau^e [X_e] : e \in \mathcal{E}^0\} \rangle^{\mathbf{S}}_{[X_{\mathcal{V}^0 \cap (\mathcal{U} \cup \mathcal{V}^1)}]}, \langle \{\tau^e [X_e] : e \in \mathcal{E}^1\} \rangle^{\mathbf{S}}_{[X_{\mathcal{V}^1 \cap (\mathcal{U} \cup \mathcal{V}^0)}]} \rangle^{\mathbf{S}}_{[X_{\mathcal{U}}]} .$$

Proof. We use the associativity and commutativity of $\otimes$ to get for any index $x_{\mathcal{V}}$

$$\bigotimes_{e \in \mathcal{E}} \tau^e [X_e = x_e] = \left(\bigotimes_{e \in \mathcal{E}^0} \tau^e [X_e = x_e] \right) \otimes \left(\bigotimes_{e \in \mathcal{E}^1} \tau^e [X_e = x_e] \right) .$$

Let us now keep $x_{\mathcal{U}}$ fixed and investigate the $\oplus$ operation over the indices $x_{\mathcal{V}\setminus\mathcal{U}}$. Distributivity of $\otimes$ over $\oplus$ implies

$$\sum_{x_{\mathcal{V}\setminus\mathcal{U}}} \bigotimes_{e \in \mathcal{E}} \tau^e [X_e = x_e] = \sum_{x_{(\mathcal{V}^0 \cap \mathcal{V}^1)\setminus\mathcal{U}}} \left(\sum_{x_{\mathcal{V}^0 \setminus (\mathcal{U} \cup \mathcal{V}^1)}} \bigotimes_{e \in \mathcal{E}^0} \tau^e [X_e = x_e] \right) \otimes \left(\sum_{x_{\mathcal{V}^1 \setminus (\mathcal{U} \cup \mathcal{V}^0)}} \bigotimes_{e \in \mathcal{E}^1} \tau^e [X_e = x_e] \right)$$

which is equivalent to

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle^{\mathbf{S}}_{[X_{\mathcal{U}} = x_{\mathcal{U}}]} = \langle \langle \{\tau^e [X_e] : e \in \mathcal{E}^0\} \rangle^{\mathbf{S}}_{[X_{\mathcal{V}^0 \cap (\mathcal{U} \cup \mathcal{V}^1)}]}, \langle \{\tau^e [X_e] : e \in \mathcal{E}^1\} \rangle^{\mathbf{S}}_{[X_{\mathcal{V}^1 \cap (\mathcal{U} \cup \mathcal{V}^0)}]} \rangle^{\mathbf{S}}_{[X_{\mathcal{U}} = x_{\mathcal{U}}]} .$$

Since therefore both tensors of the claimed contraction equation coincide on all coordinates, they are equal. □

As a consequence, message passing schemes such as variable elimination generalize to semiring tensor network contractions.

1.1.2 Examples

1.1.2.1 Sum-product TN

Standard TN use the semiring

$$(\mathbb{R}, +, \cdot, 0, 1) .$$

Variants are:

- Restriction of $\mathbb{R}$ to non-negative numbers $[0, \infty)$: Ensuring the probabilistic interpretation
- Generalization of $\mathbb{R}$ to complex numbers $\mathbb{C}$: Used in quantum computing

1.1.2.2 Boolean semiring

We take $S = \{0, 1\}$ and interpret 0 as False, 1 as True. We define the disjunction and conjunction operations by

$$\vee [X_0, X_1] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} [X_0, X_1]$$

and

$$\wedge [X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] .$$

Tensors over $\{0, 1\}$ are interpreted as constraints, and their tensor networks as constraint networks.

The satisfiability of a constraint network $\{\tau^e [X_e] : e \in \mathcal{E}\}$ is the contraction

$$\bigvee_{x_\mathcal{V}} \bigwedge_{e \in \mathcal{E}} \tau^e [X_e = x_e] = \langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle_\emptyset^{(\{0,1\}, \vee, \wedge, 0, 1)} .$$

As an additional property, we have invariance of adding subcontractions to the network. In general tensor networks, this holds only for the support of subcontractions (being the operator mapping 0 to 0 and any other $s \in S$ to 1).

The relation with the standard semiring is for boolean tensor networks

$$\langle \tau^\mathcal{G} \rangle_{[X_\mathcal{U}]}^{(\{0,1\}, \vee, \wedge, 0, 1)} \mathbb{I}_{\neq 0} \left(\langle \tau^\mathcal{G} \rangle_{[X_\mathcal{U}]} \right)$$

In this way, the message passing based algorithms in Chapter **??** and Chapter **??** simplify when chosing the satisfaction semiring instead of the default semiring.

We can use the Boolean semiring to represent first-oder logic formulas with existential quantifications as contraction equations. A formula

$$\forall x_\mathcal{U} : \exists x_\mathcal{W} : f [X_\mathcal{U} = x_\mathcal{U}, X_\mathcal{W} = x_\mathcal{W}]$$

is the contraction equation

$$\langle f [X_{\mathcal{U} \cup \mathcal{W}}] \rangle_{[X_\mathcal{U}]}^{(\{0,1\}, \vee, \wedge, 0, 1)} = \mathbb{I} [X_\mathcal{U}] .$$

1.1.2.3 Maximum calculus

The maximum semiring is the tuple

$$(\mathbb{R}^+, \max, \cdot, 0, 1) .$$

The full contraction is the maximal coordinate in the tensor

$$\langle \tau \left[X_{[d]} \right] \rangle_\emptyset^{(\mathbb{R}^+, \max, \cdot, 0, 1)} = \max_{x_{[d]} \in \bigtimes_{k \in [d]} [m_k]} \tau \left[X_{[d]} = x_{[d]} \right] .$$

When leaving variables open in the contraction, we get the maximal coordinate in the specific slices of the tensor, namely

$$\langle \tau \left[X_{[d]} \right] \rangle_{[X_{\mathcal{U}} = x_{\mathcal{U}}]}^{(\mathbb{R}^+, \max, \cdot, 0, 1)} = \max_{x_{\mathcal{V} \setminus \mathcal{U}}} \tau \left[X_{[d]} = x_{[d]} \right] .$$

Analogous properties has the minimal semiring

$$(\mathbb{R}^+, \min, \cdot, \infty, 1) .$$

These schemes only work, when taking the minimal coordinate keeping all variables open. When searching for the maximal/minimal coordinate of a contraction with some variables closed, we need to first coarse grain the tensor network.

These contractions are used in encoding-decoding schemes [9, 8]. Message passing schemes involving the maximum semiring are known as max-product algorithms [3], and generalizations of the Viterbi algorithm.

1.1.2.4 Tropical semiring

The tropical semiring is

$$\mathbb{T} = (\mathbb{R} \cup \{\infty\}, \min, +, \infty, 0) .$$

The contraction of a tensor gives the minimum

$$\langle \tau \left[X_{[d]} \right] \rangle_{\varnothing}^{\mathbb{T}} = \min_{x_{[d]} \in \bigtimes_{k \in [d]} [m_k]} \tau \left[X_{[d]} \right] .$$

The tensor product in the tropical semiring sums the leg vector coordinates, that is

$$\langle \{ \rho^k \left[X_k \right] \, : \, k \in [d] \} \rangle_{[X_{[d]} = x_{[d]}]}^{\mathbb{T}} = \sum_{k \in [d]} \rho^k \left[X_k = x_k \right] .$$

Further application in [2] and on spin glasses [7].

1.1.2.5 Galois field of two elements

The Galois field of two elements is the semiring

$$\mathbb{F}_2 := (\{0, 1\}, \oplus, \cdot, 0, 1) ,$$

where $\oplus$ denotes the sum $\mod 2$ and $\cdot$ the standard multiplication.

Example 1.1 Contractions in the semiring $\mathbb{F}_2$ While $\left\langle f \left[X_{[d]} \right] \right\rangle_{[\varnothing]}$ is the model count, the contraction with respect to $\mathbb{F}_2$ decides whether the number of models is odd or even:

$$\begin{aligned} \langle f \left[X_{[d]} \right] \rangle_{\varnothing}^{\mathbb{F}_2} &= \bigoplus_{x_{[d]} \in \bigtimes_{k \in [d]} [m_k]} f \left[X_{[d]} = x_{[d]} \right] \\ &= \left| \{ x_{[d]} \, : \, f \left[X_{[d]} = x_{[d]} \right] = 1 \} \right| \mod 2 . \end{aligned}$$

We will investigate that semiring further in Sect. **??**, where we relate CP decompositions with respect to that semiring to Reed-Muller expansions.

1.1.3 Tensor network formats

Based on the semiring contraction, we can define tensor network formats for arbitrary semirings.

- EL depends only on the multiplicative structure of the semiring. Therefore equivalent for all the semirings with the standard product.
- CP format is the semiring sum of EL. Examples in the galois field semiring in Sect. **??**.

1.1.4 Basis calculus

Basis calculus can be generalized to arbitrary semirings.

Definition 1.4 Let $\mathbf{S}$ be a commutative semiring and $q : \bigtimes_{k\in[d]}[m_k] \to \bigtimes_{\ell\in[p]}[n_\ell]$ a function. Then the basis encoding of q with respect to the $\mathbf{S}$ is the tensor $\beta^q\left[Y_{[p]}, X_{[d]}\right]$ with coordinates

$$\beta^{q,\mathbf{S}}[Y_{[p]} = y_{[p]}, X_{[d]} = x_{[d]}] = \begin{cases} \mathbf{1} & \text{if} \quad q(x_{[d]}) = y_{[p]} \\ \mathbf{0} & \text{else}\,. \end{cases}$$

Based on the restricted group operation tables we show the decomposition property of generic basis encodings.

Theorem 1.2 *Let $\mathbf{S}$ be a commutative semiring and $q : \bigtimes_{k\in[d]}[m_k] \to [n]$ and $g : [n] \to \bigtimes_{\ell\in[p]}[n_\ell]$ functions. Then we have*

$$\langle\beta^{q,\mathbf{S}}[Z, X_{[d]}], \beta^{g,\mathbf{S}}[Y_{[p]}, Z]\rangle^{\mathbf{S}}_{[X_{[d]},Y_{[p]}]} = \beta^{g\circ q,\mathbf{S}}[Y_{[p]}, X_{[d]}]$$

Proof. For any indices $x_{[d]}$ and $y_{[p]}$ we have

$$\begin{aligned}&\langle\beta^{q,\mathbf{S}}[Z, X_{[d]}], \beta^{g,\mathbf{S}}[Y_{[p]}, Z]\rangle^{\mathbf{S}}_{[X_{[d]}=x_{[d]},Y_{[p]}=y_{[p]}]} \\ &\quad= \bigoplus_{z\in[n]} \beta^{q,\mathbf{S}}[Z = z, X_{[d]} = x_{[d]}] \otimes \beta^{g,\mathbf{S}}[Y_{[p]} = y_{[p]}, Z = z]\end{aligned}$$

Now, for any z for which either $q(x_{[d]}) \neq z$ or $g(z) \neq y_{[p]}$, the summand is the neutral element $\mathbf{0}$ ($\mathbf{0}\otimes\mathbf{0}$, $\mathbf{0}\otimes\mathbf{1}$ or $\mathbf{1}\otimes\mathbf{0}$). If and only if $g \circ q(x_{[d]}) = y_{[p]}$, there is one summand $\mathbf{1}\otimes\mathbf{1} = \mathbf{1}$, and the contraction coordinate of the contraction is thus $\mathbf{1}$. The coordinates therefore coincide with those of $\beta^{g\circ q,\mathbf{S}}[Y_{[p]}, X_{[d]}]$. □

1.2 Contraction of tree tensor networks

We first define tree hypergraphs.

Definition 1.5 A hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is called connected, if for any two nodes $v, \tilde{v} \in \mathcal{V}$ we find a sequence $v_{[n]}$ of nodes, such that $v_0 = v, v_{n-1} = \tilde{v}$ and for any $k \in [n-1]$ there is $e \in \mathcal{E}$ with $v_k, v_{k+1} \in e$. A $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is called a tree, if it is connected and if for any $e \in \mathcal{E}$ the hypergraph $(\mathcal{V}, \mathcal{E}\backslash\{e\})$ is not connected.

1.2.1 Tree message passing

1.2.2 Analysis

We denote for each pair (e_0, e_1) the subset $\mathcal{E}^{\to(e_0,e_1)} \subset \mathcal{E}$ as the subset of edges $e \in \mathcal{E}$, for which each path to e_1 passes through e_0. Note, that by construction $e_0 \in \mathcal{E}^{\to(e_0,e_1)}$.

Theorem 1.3 *For any tensor network on a tree hypergraph, Algorithm* **??** *terminates in the tree-based implementation and returns final messages*

$$\chi_{e_0\to e_1}\left[X_{e_0\cap e_1}\right] = \langle\{\tau^e\left[X_e\right] : e \in \mathcal{E}^{\to(e_0,e_1)}\}\rangle^{\mathbf{S}}_{[X_{e_0\cap e_1}]}$$

Algorithm 1 Tree Belief Propagation

Require: Tensor network $\tau^{\mathcal{G}}$ on a hypergraph $\mathcal{G}$ with respect to a semiring $\mathbf{S}$
Ensure: Messages $\{\chi_{e_0\to e_1}\left[X_{e_0\cap e_1}\right] : (e_0,e_1)\in\mathcal{E}^{\to}\}$

Initialize a message scheduler $S = \{(e_0,e_1)\in\mathcal{E}^{\to} : e_0 \text{ a leaf in the tree } (\mathcal{E},\mathcal{E}^{\to})\}]$
while S not empty **do**
Pop a (e_0,e_1) pair from S
Compute the message

$$\chi_{e_0\to e_1}\left[X_{e_0\cap e_1}\right] = \langle\{\tau^{e_0}\left[X_{e_0}\right]\}\cup\{\chi_{e_2\to e_0}\left[X_{e_2\cap e_0}\right] : (e_2,e_0)\in\mathcal{E}^{\to},\ e_2\neq e_1\}\rangle^{\mathbf{S}}_{[X_{e_0\cap e_1}]}$$

Update S by all messages (e_1,e_3) which have not yet been sent, if all messages (e_2,e_1) with $e_2\neq e_3$ have been sent.
end while
return Messages $\{\chi_{e_0\to e_1}\left[X_{e_0\cap e_1}\right] : (e_0,e_1)\in\mathcal{E}^{\to}\}$

Proof. We show this property by induction over the edge sets $\mathcal{E}^{\to(e_0,e_1)}$ to pairs $(e_0,e_1)\in\mathcal{E}^{\to}$, such that $\left|\mathcal{E}^{\to(e_0,e_1)}\right|\leq n$. Notice, that since always $e_0\in\mathcal{E}^{\to(e_0,e_1)}$ we have $n\geq 1$.

$n=1$: In this case we have $\mathcal{E}^{\to(e_0,e_1)}=\{e_0\}$ and e_0 is a leaf of the tree-hypergraph $\mathcal{G}$. The claimed message property holds thus by definition.

$n\to n+1$: Let us assume, that the message obeys the claimed property for edge sets with cardinality up to n. If there is no edge set with cardinality $n+1$, the property holds also for those with cardinality up to $n+1$. If there is an edge set $\mathcal{E}^{\to(e_0,e_1)}$ with size $n+1$, we have

$$\mathcal{E}^{\to(e_0,e_1)} = \{e_0\}\cup\left(\bigcup_{e_2\in\mathcal{E}^{\to}}\mathcal{E}^{\to(e_2,e_0)}\right).$$

The message $\chi_{e_0\to e_1}$ is sent, once all messages $\chi_{e_2\to e_0}$ to $(e_2,e_0)\in\mathcal{E}^{\to}\{(e_1,e_0)\}$ arrived. By definition we have

$$\chi_{e_0\to e_1}\left[X_{e_0\cap e_1}\right] = \langle\{\tau^{e_0}\left[X_{e_0}\right]\}\cup\{\chi_{e_2\to e_0}\left[X_{e_2\cap e_0}\right] : (e_2,e_0)\in\mathcal{E}^{\to},\ e_2\neq e_1\}\rangle^{\mathbf{S}}_{[X_{e_0\cap e_1}]}$$

Now we use the induction assumption on $\mathcal{E}^{\to(e_2,e_0)}$ (since its cardinality is at most n) and get

$$\begin{aligned}
\chi_{e_0\to e_1}\left[X_{e_0\cap e_1}\right] &= \langle\{\tau^{e_0}\left[X_{e_0}\right]\}\cup\left(\bigcup_{(e_2,e_0)\in\mathcal{E}^{\to},\,e_2\neq e_1}\langle\{\tau^{e_3}\left[X_{e_3}\right] : e_3\in\mathcal{E}^{\to(e_2,e_0)}\}\rangle^{\mathbf{S}}_{[X_{e_2\cap e_0}]}\right)\rangle^{\mathbf{S}}_{[X_{e_0\cap e_1}]}\\
&= \langle\{\tau^{e_0}\left[X_{e_0}\right]\}\cup\left(\bigcup_{(e_2,e_0)\in\mathcal{E}^{\to},\,e_2\neq e_1}\{\tau^{e_3}\left[X_{e_3}\right] : e_3\in\mathcal{E}^{\to(e_2,e_0)}\}\right)\rangle^{\mathbf{S}}_{[X_{e_0\cap e_1}]}\\
&= \langle\{\tau^{e}\left[X_{e}\right] : e\in\mathcal{E}^{\to(e_0,e_1)}\}\rangle^{\mathbf{S}}_{[X_{e_0\cap e_1}]}
\end{aligned}$$

Here we used the commutation of contraction property in the second equation, which is justified by the assumed tree property of the hypergraph. Thus, the message property holds also for any edge sets of size $n+1$.

By induction, the claimed message property therefore holds for all final messages. □

1.3 Boolean message passing

We investigate message passing schemes with respect to the Boolean semiring $(\{0,1\},\vee,\wedge,0,1)$. To this end, we derive the schemes based on properties of the Boolean semiring and then show their application to sparsify generic contractions.

1.3.1 Properties of the Boolean semiring

The main property of contractions in the Boolean semiring is the invariance under adding subcontractions, which we state in the next theorem.

Theorem 1.4 *Let $\tau^{\mathcal{G}}$ be an tensor network on an arbitrary hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with respect to the Boolean semiring. For any subsets $\mathcal{U} \subset \mathcal{V}$ and $\tilde{\mathcal{E}} \subset \mathcal{E}$ we have*

$$\langle \tau^{\mathcal{G}} \cup \{\langle\{\tau^e\left[X_e\right] \,:\, e \in \tilde{\mathcal{E}}\}\rangle^{(\{0,1\},\vee,\wedge,0,1)}_{[X_{\mathcal{U}}]}\}\rangle^{(\{0,1\},\vee,\wedge,0,1)}_{[X_{\mathcal{V}}]} = \langle \tau^{\mathcal{G}}\rangle^{(\{0,1\},\vee,\wedge,0,1)}_{[X_{\mathcal{V}}]} \,.$$

Proof. Coordinatewise: Only invariant, if subcontraction evaluates to 0. But then by monotonocity of subcontractions also the whole contraction evaluates to 0, therefore invariant. □

1.3.2 Application on sum-product contractions

Thm. 1.4 can be generalized to contractions on more general semirings sharing the elements 0, 1 with $(\{0,1\}, \vee, \wedge, 0, 1)$. In such way we can switch to message passing schemes in the Boolean semiring to sparsify more generic tensor networks.

1.3.3 Generic support tensors

Even more generic, we define the support operator on a generic tensor with respect to a semiring $\mathbf{S}$ as

$$\mathbb{I}_{\neq \mathbf{0}} \circ \tau\left[X_{[d]} = x_{[d]}\right] = \begin{cases} \mathbf{0} & \text{if} \quad \tau\left[X_{[d]} = x_{[d]}\right] = \mathbf{0} \\ \mathbf{1} & \text{else}\,. \end{cases}$$

We notice that the support operator leaves any tensor over semirings with $S = \{\mathbf{0}, \mathbf{1}\}$ invariant. This is the case for the Boolean semiring $(\{0,1\}, \vee, \wedge, 0, 1)$ and the Galois field $\mathbb{F}_2$.

Lemma 1.1 *Any tensor is invariant under contraction with its support tensor, that is*

$$\tau\left[X_{[d]}\right] = \langle \tau\left[X_{[d]}\right], \mathbb{I}_{\neq \mathbf{0}} \circ \tau\left[X_{[d]}\right]\rangle^{\mathbf{S}}_{[X_{[d]}]}\,.$$

Proof. For any coordinate $s \neq \mathbf{0}$ the support operator returns $\mathbf{1}$, and by the multiplicative invariance property we have $s \otimes \mathbf{1} = s$. If $s = \mathbf{0}$ the support operator returns $\mathbf{0}$ and we have $\mathbf{0} \otimes \mathbf{0} = \mathbf{0}$. □

Definition 1.6 A semiring $\mathbf{S}$ is called *inverse-free*, if for any $s_0, s_1 \in S$ with $s_0 \oplus s_1 = \mathbf{0}$ we have $s_0 = s_1 = \mathbf{0}$.

Note that $([0,\infty), +, \cdot, 0, 1)$ is inverse-free, while $(\mathbb{R}, +, \cdot, 0, 1)$ is not. The Boolean semiring $(\{0,1\}, \vee, \wedge, 0, 1)$ is inverse-free, but the Galois field $\mathbb{F}_2$ is not.

Lemma 1.2 *The support of a contraction obeys*

$$\mathbb{I}_{\neq \mathbf{0}} \circ \langle \tau^{\mathcal{G}}\rangle^{\mathbf{S}}_{[X_{\mathcal{U}}]} \prec \langle\{\mathbb{I}_{\neq \mathbf{0}} \circ \tau^e\left[X_e \,:\, e \in \mathcal{E}\right]\}\rangle^{(\{0,1\},\vee,\wedge,0,1)}_{[X_{\mathcal{U}}]}\,.$$

Here the Boolean semiring is defined with the neutral elements $\mathbf{0}, \mathbf{1}$ of the semiring $\mathbf{S}$, and the inequality denotes partial ordering of Boolean tensors. If the semiring is inverse free, then the above inequality holds straight.

Proof. To show the first claim, let $x_{\mathcal{U}}$ be an index such that

$$\langle\{\mathbb{I}_{\neq \mathbf{0}} \circ \tau^e\left[X_e\right] \,:\, e \in \mathcal{E}\}\rangle^{(\{0,1\},\vee,\wedge,0,1)}_{[X_{\mathcal{U}} = x_{\mathcal{U}}]} = \mathbf{0}\,.$$

It suffices to show that for this index we have $\langle \tau^{\mathcal{G}}\rangle^{\mathbf{S}}_{[X_{\mathcal{U}} = x_{\mathcal{U}}]} = \mathbf{0}$. Since the Boolean semiring is inverse-free we have for any extension $x_{\mathcal{V}}$ of $x_{\mathcal{U}}$ to $\mathcal{V}$ that

$$\langle\{\mathbb{I}_{\neq \mathbf{0}} \circ \tau^e\left[X_e \,:\, e \in \mathcal{E}\right]\}\rangle^{(\{0,1\},\vee,\wedge,0,1)}_{[X_\mathcal{V}=x_\mathcal{V}]} = \mathbf{0}\,.$$

We therefore find $e \in \mathcal{E}$ such that for the restriction x_e of $x_\mathcal{V}$ to e we have

$$\tau^e\left[X_e = x_e\right] = \mathbf{0}$$

and thus

$$\langle\tau^\mathcal{G}\rangle^{\mathbf{S}}_{[X_\mathcal{V}=x_\mathcal{V}]} = \mathbf{0}\,.$$

It follows that also

$$\langle\tau^\mathcal{G}\rangle^{\mathbf{S}}_{[X_\mathcal{U}=x_\mathcal{U}]} = \mathbf{0}\,.$$

The second claim, namely the vise versa inequality, can be shown analogously using that $\mathbf{S}$ is inverse-free. □

As an example, where the inequality in Lem. 1.2 does not hold straight, consider the contraction of

$$\tau^0\left[X_0\right]\begin{pmatrix}1\\-1\end{pmatrix}\left[X_0\right] \quad \text{and} \quad \tau^1\left[X_0,X_1\right]\begin{pmatrix}1&1\\1&1\end{pmatrix}\left[X_0,X_1\right]\,.$$

The support is the trivial tensor in both cases, but $\left\langle\tau^0\left[X_0\right],\tau^1\left[X_0,X_1\right]\right\rangle_{[X_1]} = \mathbb{I}\left[X_1\right]$. When we restrict to the non-negative real numbers in the sum-product semiring, the semiring is inverse free and Lem. 1.2 would guarantee equality.

Boolean message passing can therefore be applied to sparsify generic contractions to arbitrary semirings:

- Add the support of the contraction (Lem. 1.1)
- Decompose the support into local network (Lem. 1.2)
- Perform Boolean message passing to simplify the support

1.4 Hypergraph manipulations

1.4.1 Bethe hypergraphs

Definition 1.7 (Delta tensor on generic semirings) Given a semiring $\mathbf{S} = (S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ and a collection of variables X_e of same dimension, the corresponding delta tensor $\delta\left[X_e\right]$

$$\delta\left[X_e = x_e\right] = \begin{cases}\mathbf{1} & \text{if} \quad \forall v,\tilde{v} \in e : x_v = x_{\tilde{v}}\\ \mathbf{0} & \text{else}\end{cases}\,.$$

The main manipulation operation with delta tensor is shown in the next lemma.

Lemma 1.3 *Given a tensor network $\tau^\mathcal{G}\left[X_{[d]}\right]$ with respect to a semiring $\mathbf{S} = (S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$. For any set of edges $\tilde{\mathcal{E}} \subset \mathcal{E}$ each containing a variable v define cores $\tilde{\tau}^e\left[X_{e\setminus\{v\}}, X_{v_e}\right]$ with coordinates*

$$\tilde{\tau}^e\left[X_{e\setminus\{v\}} = x_{e\setminus\{v\}}, X_{v_e} = x_v\right] = \tau^e\left[X_e = x_e\right]\,.$$

Then we have

$$\tau^\mathcal{G}\left[X_{[d]}\right] = \langle\{\tau^e\left[X_e\right] \,:\, e \notin \tilde{\mathcal{E}}\} \cup \{\tilde{\tau}^e\left[X_{e\setminus\{v\}}, X_{v_e}\right] \,:\, e \in \tilde{\mathcal{E}}\} \cup \{\delta\left[X_v, X_{v_e} \,:\, e \in \tilde{\mathcal{E}}\right]\}\rangle^{\mathbf{S}}_{[X_\mathcal{V}]}\,.$$

Proof. Using the neutrality properties of $\mathbf{1}$ and $\mathbf{0}$. □

Definition 1.8 Given a tensor network $\tau^\mathcal{G}$ on a decorated hypergraph $\mathcal{G}$, we define the Bethe hypergraph $\tilde{\mathcal{G}}$ as $(\mathcal{U}, \tilde{\mathcal{E}} \cup \Delta^\mathcal{G})$ where we have

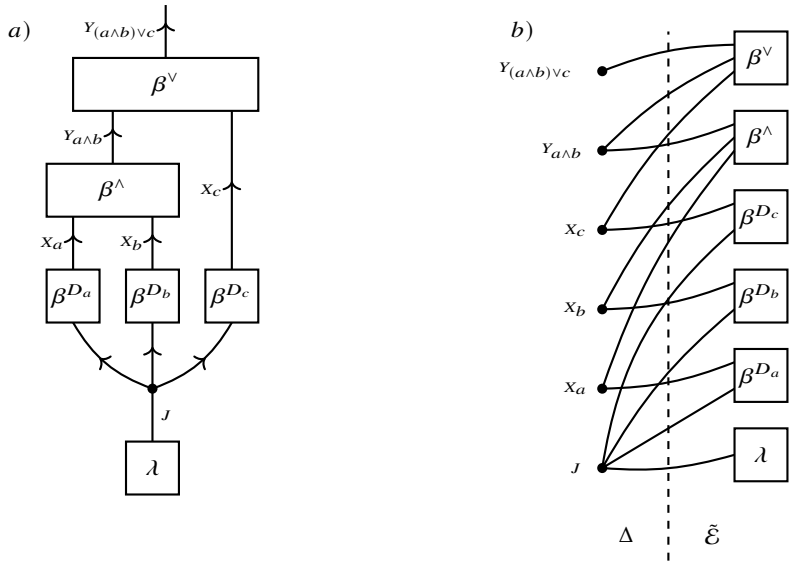


Fig. 1.1 Example of a Bethe Cluster Graph. a) Example of a Tensor Network $\tau^{\mathcal{G}}$, which represents the by λ averaged evaluation of the formula $(a \wedge b) \vee c$ on data D. b) Corresponding Bethe Cluster Hypergraph, which dual is bipartite by the sets Δ and $\tilde{\mathcal{E}}$.

- Recolored Edges $\tilde{\mathcal{E}} = \{\tilde{e} \,:\, e \in \mathcal{E}\}$ where $\tilde{e} = \{v^e \,:\, v \in e\}$, which decoration tensor has same coordinates as τ^e
- Nodes $\mathcal{U} = \mathcal{V} \cup (\bigcup_{e\in\mathcal{E}} \tilde{e})$
- Delta Edges $\Delta^{\mathcal{G}} = \left\{\{v\} \cup \{v^e \,:\, e \ni v\} \,:\, v \in \mathcal{V}\right\}$, each of which decorated by a delta tensor $\delta^{\{v^e : e\ni v\}}$

The resulting overlap graph is bipartite, since any variable appears exactly in one cluster in $\tilde{\mathcal{E}}$ and in one cluster of $\Delta^{\mathcal{G}}$. This further makes the dual of the Bethe Cluster Hypergraph a proper graph (i.e. edges consistent of node pairs).

We effectively copy each node v by v^e. Let us now show, that when assigning dirac delta tensors the contraction reproduces the original contraction.

By adding delta tensors to each node $v \in \mathcal{V}$ and defining its leg variables by v^e for $e \in \mathcal{E}$. We mark each such delta tensor by a cluster in $\Delta^{\mathcal{G}}$, as defined in the following (see also Figure 1.1).

By Lem. 1.4 this construction does not change contractions.

Lemma 1.4 (Bethe Invariance) *Given a tensor network $\tau^{\mathcal{G}}\left[X_{[d]}\right]$ on a hypergraph $\mathcal{G}$. We define a tensor network on its bethe hypergraph $\tilde{\mathcal{G}}$ by the tensors for $e \in \mathcal{E}$*

$$\tilde{\tau}^{\tilde{e}}\left[X_{\tilde{e}}\right] = \left\langle \tau^{e}\left[X_e\right], \delta\left[X_e, X_{\tilde{e}}\right]\right\rangle_{[X_{\tilde{e}}]}$$

and the tensors for $\tilde{e} \in \Delta^{\mathcal{G}}$

$$\tilde{\tau}^{\tilde{e}}\left[X_{\tilde{e}}, X_v\right] = \delta\left[X_{\tilde{e}}, X_v\right] .$$

Then we have

$$\tau^{\mathcal{G}}\left[X_{[d]}\right] = \left\langle \{\tilde{\tau}^{\tilde{e}}\left[X_{\tilde{e}}\right] \,:\, \tilde{e} \in \tilde{\mathcal{E}} \cup \Delta^{\mathcal{G}}\}\right\rangle_{[X_{\mathcal{V}}]}$$

Proof. By invariance of adding dirac deltas. □

Equal constructions:

- BP on factor graphs in [9]
- Bethe Cluster graphs in [3]

1.4.2 Coarse graining

We often account for tensor networks, which are not defined on a tree hypergraph. In that case, one can perform graph manipulations, which we call coarse graining schemes, to reformulate the contraction on an equivalent tree hypergraph.

Definition 1.9 Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and $\tilde{\mathcal{G}} = (v, \tilde{\mathcal{E}})$ be two hypergraphs sharing their node set and $c : \mathcal{E} \to \tilde{\mathcal{E}}$ a map. We say that c is a coarse graining map and $\tilde{\mathcal{G}}$, if for all $e \in \mathcal{E}$

$$e \subset c(e)\,.$$

For any tensor network $\tau^{\mathcal{G}}\,[X_{\mathcal{V}}]$ on $\mathcal{G}$ we further define a coarse grained tensor network $\tilde{\tau}^{\tilde{\mathcal{G}}}\,[X_{\mathcal{V}}]$ on $\tilde{\mathcal{G}}$ with hypercores to $\tilde{e} \in \tilde{\mathcal{E}}$ by

$$\tilde{\tau}^{\tilde{e}}\,[X_{\tilde{e}}] = \langle \{\tau^{e}\,[X_e]\ :\ \tilde{e} = c(e)\}\rangle^{\mathbf{S}}_{[X_{\tilde{e}}]}\,.$$

In the literature such coarse graining procedures appear as Cluster Graphs.

We notice that coarse graining of tensor networks preserves their contractions, since

$$\begin{aligned}
\tau^{\mathcal{G}}\,[X_{\mathcal{V}}] &= \langle \{\tau^{e}\,[X_e]\ :\ e \in \mathcal{E}\}\rangle^{\mathbf{S}}_{[X_{\mathcal{V}}]} \\
&= \langle \bigcup_{\tilde{e}\in\tilde{\mathcal{E}}} \{\tau^{e}\,[X_e]\ :\ c(e) = \tilde{e}\}\rangle^{\mathbf{S}}_{[X_{\mathcal{V}}]} \\
&= \langle \{\langle \{\tau^{e}\,[X_e]\ :\ c(e) = \tilde{e}\}\rangle^{\mathbf{S}}_{[X_e]}\ :\ \tilde{e} \in \tilde{\mathcal{E}}\}\rangle^{\mathbf{S}}_{[X_{\mathcal{V}}]} \\
&= \langle \{\tilde{\tau}^{\tilde{e}}\,[X_{\tilde{e}}]\ :\ \tilde{e} \in \tilde{\mathcal{E}}\}\rangle^{\mathbf{S}}_{[X_{\mathcal{V}}]} \\
&= \tilde{\tau}^{\tilde{\mathcal{G}}}\,[X_{\mathcal{V}}]\ .
\end{aligned}$$

1.4.2.1 Variable Elimination

Given an order of the nodes the variable elimination procedure provides a coarse graining scheme of an augmented graph.

The variable elimination algorithm is simultaneously defining an augmentation of a tensor network with delta cores and defining a coarse graining mechanism of the augmentation.

Lemma 1.5 *The Algorithm 2 returns a coarse graining map c towards a tree hypergraph.*

Proof. By construction we have $e \subset c(e)$ for any $e \in \mathcal{E}$, thus c is a coarse graining map.

□

1.4.2.2 Junction Tree Algorithm

Classically the junction tree algorithm is formulated on the extended graph (we define extended graphs here by two nodes adjacent if and only if they are contained in a hypergraph) [4], where one exploits that a graph has a junction tree if and only if it is triangulated [5]. The junction tree algorithm therefore constructs a triangulation of the graph by adding edges (also called chordal graph [3]), and gets a tree-hypergraph by its cliques. The coarse graining map to a tree hypergraph is thus the selection of the maximal clique in the triangulated extended graph, which contains the nodes of the hyperedge. Note that finding a minimal triangulation is NP-hard.

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Algorithm 2 Variable Elimination

Require: Tensor network $\{\tau^e\,[X_e]\,:\,e\in\mathcal{E}\}$ with total order of the nodes
Ensure: Coarse graining map c towards a tree hypergraph $(\mathcal{V},\tilde{\mathcal{E}})$

$\mathcal{C}\leftarrow\emptyset$ of cores already contracted
$c:\emptyset\to\emptyset$
$\mathcal{M}$ auxiliary edges (messages)
for $v\in\mathcal{V}$ following the total order **do**
Step 1: Plan new message e_v given the current ones $\mathcal{M}$

- Choose all edges from $\mathcal{E}$ containing v, which have not been contracted before and update the contracted ones

$$\mathcal{E}^v\leftarrow\{e\,:\,e\in\mathcal{E}\backslash\mathcal{C},\,v\in e\}$$
$$\mathcal{C}\leftarrow\mathcal{C}\cup\mathcal{E}^v$$

- Choose and remove all message edges from $\mathcal{M}$ containing v

$$\mathcal{M}^v\leftarrow\{e_{\tilde{v}}\,:\,e_{\tilde{v}}\in\mathcal{M},\,v_{\tilde{v}}\in e_{\tilde{v}}\}$$
$$\mathcal{M}\leftarrow\mathcal{M}\backslash\mathcal{M}^v$$

- Build a coarse grained edge and add it to the messages

$$e_v\leftarrow\{v\}\cup\bigcup_{e\in\mathcal{E}^v}\{\bar{v}^v\,:\,\bar{v}\in e\backslash\{v\}\}\cup\bigcup_{e_{\tilde{v}}\in\mathcal{M}^v}\{\bar{v}^{\tilde{v}}\,:\,\bar{v}^{\tilde{v}}\in e_{\tilde{v}}\backslash\{\tilde{v}\}\}$$
$$\mathcal{M}\leftarrow\mathcal{M}\cup\{e_v\}$$

Step 2: Build manipulated cores of an augmented network, which are coarse grained to e_v

- Rename the variables at each $e\in\mathcal{E}^v$, keeping only v

$$\tilde{\tau}^e\left[X_v,X_{\bar{v}_v}\,:\,\tilde{v}\in e\backslash\{v\}\right]\leftarrow\langle\tau^e\,[X_e]\cup\{\delta\left[X_{\tilde{v}},X_{\bar{v}_v}\right]\,:\,\tilde{v}\in e\backslash\{v\}\}\rangle^{\mathbf{S}}_{[X_v,X_{\bar{v}_v}\,:\,\bar{v}\in e\backslash\{v\}]}$$

- Add delta core to current variable

$$\delta\left[X_v,X_{v_{\tilde{v}}}\,:\,e_{\tilde{v}}\in\mathcal{M}^v\right]$$
$$c(\{v\}\cup\{v_{\tilde{v}}\,:\,e_{\tilde{v}}\in\mathcal{M}^v\})=e_v$$

- Add delta cores to any other variable $\bar{v}^v\in e_v\{v\}$

$$\delta\,[X_{\bar{v}^v},X_{\tilde{v}^{\bar{v}}}\,:\,e_{\tilde{v}}\in\mathcal{M}^v,\,\bar{v}\in\tilde{v}]\,c(\{\bar{v}^v\}\cup\{\bar{v}^{\tilde{v}}\,:\,e_{\tilde{v}}\in\mathcal{M}^v,\,\bar{v}\in\tilde{v}\})=e_v$$

end for
return $c,\tilde{\mathcal{E}}$

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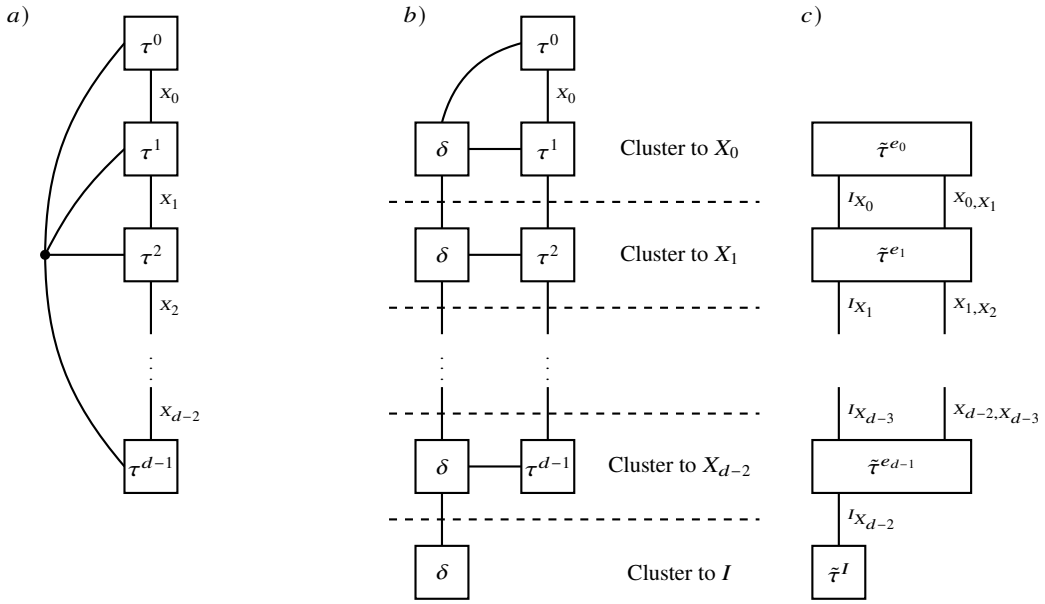


Fig. 1.2 Variable Elimination on a tensor network in the order $(X_0, \ldots, X_{d-1}, I)$. a) Original contraction on a non-tree tensor network, b) Coarse grained Bethe hypergraph and c) Coarse grained contraction.